

FACULTY OF ENGINEERING & TECHNOLOGY

Third Year Bachelor of Technology

Course Code: 202060621

Course Title: Multimedia Systems and Applications

Assignment 2

Sr. No	Question Detail
1	Discuss key techniques used in video compression, including spatial redundancy reduction, temporal redundancy reduction, and inter-frame prediction.
2	Explain the basic architecture of MPEG video compression.
3	Discuss inter-frame and intra-frame coding in MPEG.
4	Differentiate between MPEG-1, MPEG-2, and MPEG-4 standards.
5	What is proxy caching? Discuss different Proxy Caching Algorithms in details.
6	Explain the role of Content Delivery Networks (CDNs) in improving multimedia content distribution.
7	Discuss techniques such as error control, handoff management, and bandwidth adaptation for enhancing multimedia delivery over wireless networks.
8	Discuss different strategies for offloading computation tasks from mobile devices to cloud servers.
9	Discuss techniques for dynamic resource provisioning, load balancing, and autoscaling in multimedia cloud services.
10	Describe the various service models of cloud computing and explain their functions.
11	Discuss essential characteristics in cloud computing.
12	Discuss QoS requirements for multimedia networking.
13	Discuss challenges in mobile multimedia communication.
14	Differentiate between live streaming and on-demand streaming.
15	Explain different classes of multimedia applications with suitable examples.
16	Describe the various types of multimedia systems.
17	Explain the key components of a multimedia system.
18	What is Virtual Reality (VR)? Explain the architecture of a Virtual Reality system.